Weekly Progress Report

Name:

Domain: Data Science / Machine Learning

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PROJECT 9

Smart City Traffic Forecasting

Traffic Pattern Analysis & Predictive Modeling

Four Junction Study • Nov 2015 – Oct 2017

Total Records	Junctions	Forecast Period	Model MAE
48,120	**4**	**4 months**	**~12.5**
training data	city intersections	Jul–Oct 2017	vehicles/hr CV

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Government Smart City Initiative • 2017

1. Executive Summary

This report presents a comprehensive analysis of traffic patterns across four key junctions in the city, as part of the Smart City transformation initiative. Using 20 months of hourly traffic data (November 2015 to June 2017), a predictive model was developed and validated to forecast vehicle counts for the subsequent four-month period (July to October 2017).

The analysis reveals significant variation in traffic load across junctions, with Junction 1 handling over three times more vehicles per hour than any other junction. The forecasting model — a Gradient Boosting Machine — achieved a cross-validated Mean Absolute Error of approximately 12.5 vehicles per hour, providing actionable and reliable predictions for infrastructure planning.

Key Findings

Finding	Detail
Busiest junction	Junction 1 — 45 vehicles/hr average, peak 156 vehicles/hr
Quietest junction	Junction 4 — 7 vehicles/hr average, operational since mid-2016
Weekend effect	Junctions 1 & 2 drop 30–35% on weekends; Junction 3 shows no weekend dip
Peak hours	11:00–12:00 and 19:00–20:00 across most junctions
Seasonal trend	Traffic dips in Nov–Jan and July; peaks in May–June
Traffic growth	Year-on-year growth evident; year is the 2nd most important model feature (19%)

2. Project Background

The government is undertaking a large-scale Smart City transformation to convert urban areas into digital, intelligent cities. A core component of this vision is an efficient and responsive traffic management system that can anticipate demand rather than react to it.

Traffic congestion causes measurable harm to citizens: longer commutes, increased pollution, and reduced emergency response effectiveness. By forecasting traffic peaks accurately, the city can:

- Dynamically adjust signal timing to reduce bottlenecks
- Schedule road maintenance during predicted low-traffic windows
- Plan infrastructure investments (new lanes, flyovers) based on future demand
- Optimize public transport deployment in parallel with private vehicle patterns

This project focuses on four city junctions, each representing a different traffic archetype, to build a generalisable forecasting framework.

3. Data Overview

3.1 Dataset Description

The dataset consists of hourly vehicle count records across four junctions. Each record contains the DateTime, Junction identifier, vehicle count, and a unique ID.

Parameter	Training Set	Test Set	Total
Date range	Nov 2015 – Jun 2017	Jul 2017 – Oct 2017	Nov 2015 – Oct 2017
Records	48,120	11,808	59,928
Junctions	1, 2, 3, 4	1, 2, 3, 4	4
Missing values	0	0 (target)	0
Granularity	1 hour	1 hour	1 hour

3.2 Junction Characteristics

Junction	Avg (veh/hr)	Std Dev	Min	Max	Records
Junction 1	45.1	23.0	5	156	14,592
Junction 2	14.3	7.4	1	48	14,592
Junction 3	13.7	10.4	1	180	14,592
Junction 4	7.3	3.5	1	36	4,344

Note: Junction 4 has fewer records (4,344 vs 14,592) indicating it became operational in mid-2016, approximately 7 months after the other junctions.

4. Traffic Pattern Analysis

4.1 Hourly Patterns

Analysis of hourly vehicle counts reveals distinct intra-day traffic rhythms for each junction.

Junction 1 — Dual-peak pattern: traffic builds from a morning low (~25 vehicles/hr at 05:00) to a midday peak (~57 vehicles/hr at 12:00), dips slightly in the afternoon, and surges again in the evening (~59 vehicles/hr at 19:00–20:00). This is typical of a mixed-use arterial road serving both office commuters and leisure traffic.

Junctions 2 & 3 — Single afternoon-evening peak: both follow a gentler curve, with the highest counts from 19:00–21:00. They experience less pronounced morning rush, suggesting limited office-commuter traffic.

Junction 4 — Flat midday profile: traffic peaks around noon (10 vehicles/hr) and remains low throughout, consistent with a secondary or residential road.

4.2 Day-of-Week Patterns

- Junctions 1 & 2 show a clear weekday-weekend split: Mon–Thu average ~50 and ~16 vehicles/hr respectively, dropping to ~33 and ~10 on weekends (−30–35%).
- Junction 3 is anomalous: weekend traffic (14.6 vehicles/hr on Saturday) is equal to or slightly above weekday levels. This suggests the junction serves a commercial or entertainment district.
- Junction 4 follows a mild weekday preference with the lowest counts on Sundays (~6 vehicles/hr).

4.3 Monthly & Seasonal Patterns

- All junctions dip in November–January (winter, lower footfall, fewer events).
- Junctions 1 and 2 peak in May–June, coinciding with pre-summer activity and outdoor events.
- A mid-year dip is visible in July across most junctions, consistent with school holidays reducing commuter traffic.
- Junction 3 shows its peak in April (18.8 vehicles/hr) rather than May–June, suggesting a different event or seasonal driver.
- Junction 4 data is only available from mid-2016, so seasonal trends are partially observable.

5. Forecasting Methodology

5.1 Feature Engineering

Raw datetime information was decomposed into a rich set of temporal features to allow the model to learn cyclical and trend-based patterns:

Feature Category	Features
Calendar	Hour, Day, Month, Year, Day-of-week, Day-of-year, Week-of-year, Quarter
Cyclical encoding	Sin/Cos transforms for hour (24-period), day-of-week (7-period), month (12-period)
Traffic flags	Is_weekend, Is_peak_hour (07–09 & 17–19), Is_night (before 06:00 or after 22:00)
Identity	Junction ID (categorical)

5.2 Model Selection

A Gradient Boosting Machine (GBM) was selected as the forecasting model for the following reasons:

- Handles non-linear interactions between time features and junction identity without manual specification
- Robust to outliers (e.g., accident days, events) compared to linear models
- Captures the year-over-year growth trend alongside cyclical patterns simultaneously
- No stationarity requirement unlike ARIMA/SARIMA models

Model hyperparameters: 300 estimators, max depth 5, learning rate 0.05, subsample 0.8. A 5-fold cross-validation was used to estimate generalisation performance.

5.3 Model Performance

Metric	Value	Interpretation
CV MAE (mean)	12.46 vehicles/hr	Average prediction error
CV MAE (std)	±8.20 vehicles/hr	Variance across folds
Training samples	48,120 records	Nov 2015 – Jun 2017
Validation method	5-fold cross-validation	Temporally stratified

5.4 Feature Importance

The model identified the following features as most predictive of vehicle counts:

Rank	Feature	Importance	Significance
1	Junction ID	**55.0%**	Each junction has a fundamentally different baseline volume
2	Year	**18.7%**	Captures year-on-year traffic growth trend
3	Hour of day	**9.3%**	Intra-day rush hour vs. off-peak variation
4	Day of year	**5.3%**	Seasonal trends and special date effects
5	Cyclical hour (sin)	**3.2%**	Smooth cyclical representation of time of day

6. Forecast Results (Jul–Oct 2017)

6.1 Predicted Monthly Averages

The model forecasts continued traffic growth through the Jul–Oct 2017 period, consistent with the upward year-on-year trend observed in historical data.

Junction	July 2017	August 2017	September 2017	October 2017
Junction 1	68.2	72.1	78.5	81.3
Junction 2	18.4	19.2	20.7	21.4
Junction 3	17.1	18.4	19.9	21.2
Junction 4	9.8	10.1	10.6	11.0

All values are average predicted vehicle counts per hour. Junction 1 is forecast to exceed 80 vehicles/hr by October 2017, representing a ~54% increase over its historical average of 45 vehicles/hr.

6.2 Infrastructure Implications

- Junction 1 requires priority attention: forecast volumes exceed historical peaks and signal a need for capacity assessment, including lane expansion or signal optimisation.
- Junctions 2 and 3 show moderate growth and may benefit from adaptive signal timing upgrades in the medium term.
- Junction 4, while low in absolute volume, shows consistent growth from a late start and should be monitored for acceleration.

- The July dip observed historically may be less pronounced in 2017 given the overall growth trend.

7. Recommendations

7.1 Immediate Actions (0–3 months)

1. Deploy adaptive traffic signal control at Junction 1 to manage the predicted 80+ vehicles/hr load.
2. Integrate forecast data into the central traffic management system for real-time signal adjustment.
3. Schedule road maintenance at all four junctions for November–January, leveraging the seasonal traffic dip.

7.2 Medium-Term Planning (3–12 months)

- Commission a capacity study for Junction 1 to evaluate lane expansion and turning bay requirements.
- Investigate Junction 3's weekend traffic anomaly to understand whether it indicates commercial activity that warrants dedicated weekend traffic management.
- Expand sensor coverage to additional junctions to build a city-wide traffic intelligence network.

7.3 Long-Term Strategy

- Incorporate external data sources (weather, events calendar, public holidays) to improve forecast accuracy during irregular periods.
- Transition to a real-time retraining pipeline so the model continuously updates as new data arrives.
- Use traffic forecasts to inform public transport scheduling, reducing private vehicle dependency at peak times.

8. Conclusion

This study has delivered a functional, validated traffic forecasting system for the four monitored city junctions. The analysis confirms substantial heterogeneity across junctions — in volume, peak timing, and weekend behaviour — underscoring the need for junction-specific rather than one-size-fits-all traffic management strategies.

The Gradient Boosting model successfully captures the key drivers of traffic variation: junction identity, long-term growth trends, time-of-day patterns, and day-of-week effects. With a cross-validated MAE of approximately 12.5 vehicles per hour, the model provides a reliable foundation for both operational traffic management and strategic infrastructure planning.

The forecast for July–October 2017 signals continued and accelerating growth at Junction 1, demanding near-term infrastructure attention. By acting on these findings, the city will be better equipped to deliver on its Smart City vision: seamless, data-driven urban mobility for its citizens.

"Include any additional comments or observations regarding overall progress, collaboration, or notable experiences during the week."